

Machine-Learning-Assisted PSA Simulation Analysis

CO₂ Capture from 1000 LHS Samples and Final-Cycle Profiles

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SEEN5320 Final Project

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Research question

Can machine learning turn a PSA simulation campaign into accurate surrogates and interpretable operating guidance?

- System: four-step/extended-cycle PSA CO₂/N₂ separation with Zeolite 13X.
- Data: 1000 successful LHS samples and final-cycle time-space profiles.
- ML tasks: regression, high-performance classification, PCA/K-means regime discovery.
- Goal: reduce simulator dependence while preserving physical interpretation.

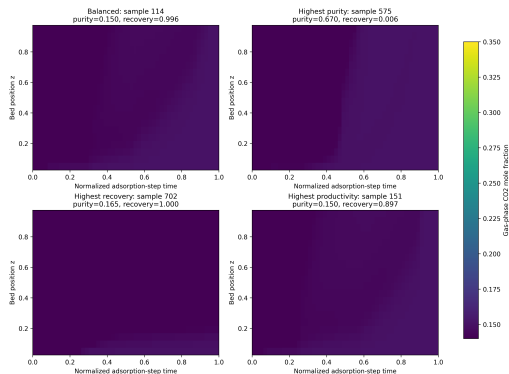
Dataset

Item	Value
Successful simulations	1000
Design variables	7
Performance KPIs	4
Profile fields	P, y, q, T, T_w
Profile features	470
Pareto samples	138



From Profiles to ML Features

- Each CSV stores the last returned cycle over time and bed position.
- For each PSA step, features summarize means, ranges, final inlet/outlet values, axial gradients, loading selectivity proxies, and gas-wall temperature lag.
- These features connect scalar KPIs to the internal state of the bed.



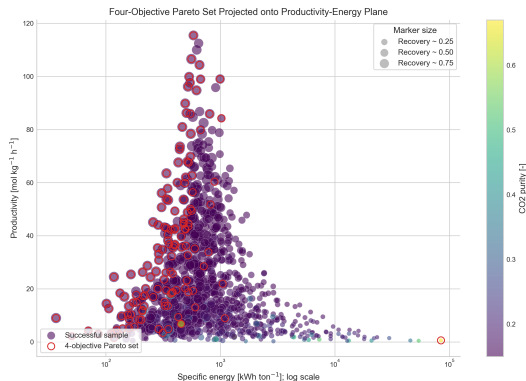
Supervised learning

- Regression targets: purity, recovery, productivity, $\log_{10}(\text{energy})$.
- Models: Ridge, random forest, gradient boosting, SVR, MLP.
- Validation: fixed 80/20 train-test split.

Screening and interpretation

- Classify top 20% balanced-score samples.
- Compute four-objective Pareto set.
- Use PCA/K-means to identify operating regimes.

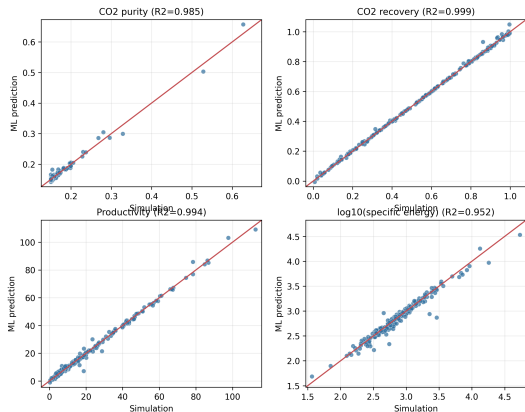
Key Tradeoff: Purity vs Recovery



- Spearman correlation between purity and recovery: -0.691 .
- Highest purity: 0.670, but recovery is only 0.0056 and energy is $84379 \text{ kWh ton}^{-1}$.
- Best balanced case: recovery 0.996 and productivity $106.6 \text{ mol kg}^{-1} \text{h}^{-1}$, but purity remains near feed level.
- This is the central process limitation in the sampled domain.

Surrogate Accuracy

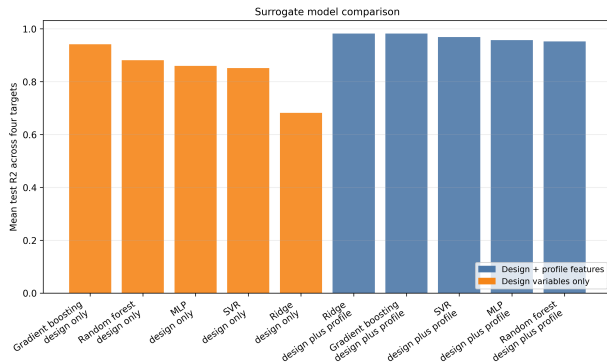
Best surrogate validation: Ridge using design plus profile



Target	RMSE	R^2
Purity	0.00590	0.985
Recovery	0.00935	0.999
Productivity	1.703	0.994
$\log_{10}(\text{energy})$	0.0897	0.952

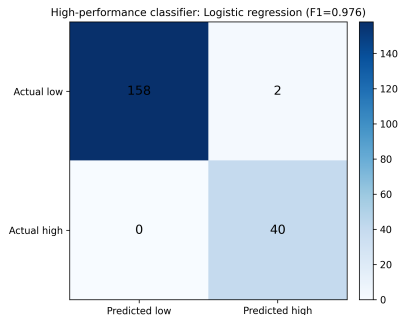
- Best model: Ridge.
- Best feature set: design + profile features.
- Mean test R^2 : 0.982.

Model Comparison



- Profile features improve every model family.
- Linear Ridge works best because final-cycle profiles already encode much of the nonlinear simulator response.
- Design-only models are weaker, especially for energy and purity.

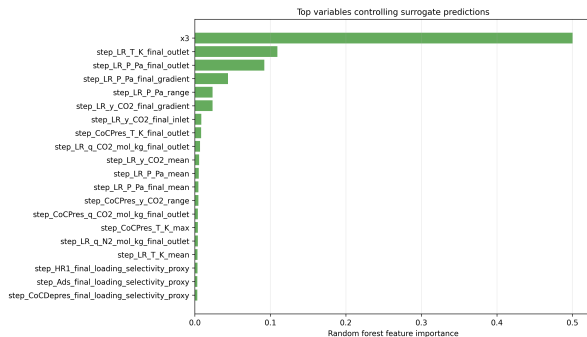
High-Performance Screening



Classifier	Accuracy	F1	ROC-AUC
Logistic reg.	0.990	0.976	0.9995
Random forest	0.950	0.865	0.9903
SVM	0.925	0.839	0.9816

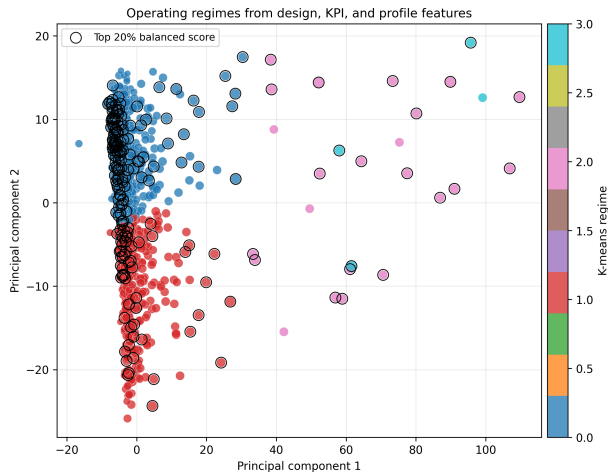
- Positive class: top 20% by balanced score.
- The high-performance region is highly separable with profile-informed features.

What Controls the Surrogate?



- Dominant variable:
light-product reflux ratio x_3 .
- Late light-reflux outlet
temperature and pressure are
also important.
- Final pressure and CO₂
gradients encode how
regenerated the bed is before
the next cycle.

Operating Regimes



- PCA uses design, KPI, and profile features.
- K-means separates broad low-to-medium performance regions from small high-recovery pockets.
- High balanced-score samples concentrate in limited zones instead of spanning the whole design space.

- ML surrogates can reproduce PSA simulation outputs accurately from one completed LHS campaign.
- Final-cycle profile features are valuable because they expose internal bed states that design variables alone miss.
- The sampled design space shows a severe purity-recovery-energy conflict.
- Practical next step: expand the domain with material variables and optimize around purity constraints, not only balanced KPI score.

Generative AI was used to assist coding, figure generation, and English drafting.

Thank you

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